ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

- **English Name: Quenton Edgar**
- **Chinese Name: 安謙宇**
- **Email: t15902105@ntu.edu.tw**
- **Student ID: t15902105**
- **University: National Taiwan University (NTU)**

Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.
4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)
- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. There are 594 sci.med documents in the data frame X.
2. Isolating categories of a dataset is an important first step because it provides insight into the distribution of the data as well as the breadth of the different categories (dimensionality), which can impact the algorithm chosen for pattern mining. It can also reveal smaller subset categories or outliers that may be overlooked or alter the outcome of a large pattern mining algorithm.

AI used for this answer: Yes / **No**

If yes, how: N/A

External references (links): None

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer: Because .dropna() removes NaN or None values from a data frame, then using the method on a dataset with 40% of the documents missing a category_name label would serve to remove 40% of the data. This would strip away valuable data that could be used for data mining. It could have a significant effect on the data, and introduce selection bias into the dataset.

A possible alternative for this would be to add data labels to the documents without a label, using "Unknown" or a similar indicator for the label, to show that the source is uncertain. This would preserve the text data while allowing algorithmic approaches to still work, despite the missing data.

AI used for this answer: Yes / **No**

If yes, how: N/A

External references (links): https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.dropna.html

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Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

1. The proportions of the original distribution do not exactly match the proportions of the sampled distribution, however the two distributions are approximately similar.
2. The risk of random sampling on a highly imbalanced dataset would be that the sample is not accurately indicative of the actual dataset due to over representation of data in the majority class and under representation of data in the minority class. Smaller subgroups may be overlooked in a random sample.

AI used for this answer: Yes / **No**

If yes, how: N/A

External references (links): https://www.investopedia.com/ask/answers/042815/what-are-disadvantages-using-simple-random-sample-approximate-larger-population.asp

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Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

English (2257, 35482)

None (2257, 35788)

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

1. The total number of features increases from 35,482 to 35,788.

2. Adding stop words back in decreases the overall sparsity of the matrix because more data is filled in with non-zero numbers to count these stop words.

AI used for this answer: Yes / No

If yes, how: N/A

External references (links):

Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

Threshold	**Number of terms kept (keep == True)**
0.0	35482
0.01	5089
0.05	1491

1. When the variance threshold changes from 0.0 to 0.05, the total vocabulary size decreases from 35,482 to 1,491 (removing 33991 of the words when the threshold is 0.05).

2. Words that have a variance close to 0 are words with extremely high document frequency or low document frequency. High document frequency words appear very commonly and appear very uniformly, such as “the” or “and”. Low document frequency words are very rare and uncommonly used, so their frequency is close to 0%.

3. Eliminating near-zero variance features is a safe way to reduce dimensionality in a dataset because it drops data that is unlikely to provide usable or significant insights. It is often attributed to noise or malformed inputs, rather than actual data, and may cause models to become distracted or create specific individual structures.

AI used for this answer: Yes / No

If yes, how: Prompt: what is text variance in data mining

External references (links):

Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what **Pearson** correlation measures versus what **Spearman** rank correlation measures?
3. In what specific scenario would a word have a high **Spearman** rank correlation with a topic, but a significantly lower **Pearson** correlation?

Answer:

1. The top-correlated words are domain-specific terms rather than generic stop words. This is because in a text about a specific topic or domain is more likely to have high correlation with domain-specific terms. Generic stop words are evenly distributed through both comp.graphics documents as well as all other documents. However, the domain-specific terms have a much higher correlation with the comp.graphics documents.
2. The Pearson correlation coefficient identifies the level of correlation between two variables, whereas Spearman correlation measures how monotonic the ordinal-ranked variables are. Pearson (r) is the covariance of X and Y divided by the product of the standard deviations and Spearman (ρ) uses the rank-distance formula.
3. If a word had a high Spearman rank correlation with a topic but had a low Pearson correlation, then the data would need to be non-linear but monotonic. This means the word's frequency rate of change is inconsistent (non-linear) resulting in a low Pearson correlation. In order to satisfy a high Spearman correlation (monotonic), the word frequency would need also need to increase as the category intensity increases.

AI used for this answer: Yes / No

If yes, how: Prompt: explain high spearman correlation in word document frequency

External references (links):

https://www.surveymonkey.com/market-research/resources/pearson-correlation-vs-spearman-correlation/

https://statistics.laerd.com/statistical-guides/spearmans-rank-order-correlation-statistical-guide.php

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Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?
3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

When minSup = 3, there are 9547 patterns generated with a runtime of ~2.6448 seconds. When minSup = 9, there are 710 patterns generated with a runtime of ~ 0.0598 seconds. Decreasing minSup significantly increases the runtime, as well as the number of discovered patterns.

Answer (2) — Top-K k vs MaxFPGrowth minSup:

The parameter k in Top-K FAE limits the output of Top-K FAE to the top k highest frequency patterns. It will always return k patterns. In MaxFPGrowth, minSup is used as a items threshold, where a data item must meet at least minSup frequency in order to be considered for pattern discovery. Unlike k, minSup does not directly determine how many patterns are returned and only sets a baseline for the frequency the item must appear at.

Answer (3) — Maximal vs standard frequent patterns:

Maximal frequent patterns are the highest frequency patterns that cannot have more items added without changing the minSup. Standard frequent patterns are all patterns that fulfill a given minSup. Because standard frequent patterns can be a large set of many patterns, maximal frequent patterns are more concise and reduce redundancy by returning a smaller pattern set.

AI used for this answer: Yes / **No**

If yes, how: N/A

External references (links):

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1. It is impossible to plot the raw X_counts matrix directly using a 2D graph. Because the columns represent every word in the dataset, so each document is a coordinate point inside a many dimensional space. Because of the high dimensionality, the raw matrix cannot be plotted on a 2D space and will make the graph clustered and visually uncertain. Additionally, the raw matrix primarily consists of zero values, because each document only uses a small subset of all unique words. This high sparsity means that the data points would collapse along the edges of the graph.
2. The category of algorithms that must be used first are dimensionality reduction algorithms found in the visualization_helpers.py file, specifically using the PAMI library.

AI used for this answer: Yes / No

If yes, how:

External references (links): https://www.ibm.com/think/topics/dimensionality-reduction

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Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. t-SNE and UMAP both show distinctive cluster islands for the document categories, while PCA shows heavy overlap around the center.
2. PCA (Principle Component Analysis) finds data of the highest variance, and does not account for topic/category as much as other factors. t-SNE and UMAP both preserve relationships between neighboring data or global data, and so clusters can be easily

observed in a lower dimensional plot. PCA, however, does not look at class labels or neighboring structures.

3. In a t-SNE plot, the local data neighborhoods are preserved but the global geometry is not preserved as well. A cluster can be near or far from another cluster, without significant meaning for specific distance. As such, data analysts cannot rely too much on distances between clusters in a t-SNE plot.

AI used for this answer: Yes / **No**

If yes, how:

External references (links): https://biostatsquid.com/pca-umap-tsne-comparison/

Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?
3. How does one-hot binarization eliminate this false assumption?

Answer:

1. 4 binary columns are created to represent the 4 categories.
2. Encoding categories as raw integer columns implies ranking and order, where one category can be ranked relative to another. It also implies the distance between the categories is significant as a unit, where there can be a larger difference between two arbitrary categories than between two other arbitrary categories. The category data is nominal, but assigning integer columns creates the impression that it is ordinal or interval data.
3. One-hot binarization replaces a column with a binary column, per category, meaning that each category has it's own vector. Because there is no order or rank to this approach, and each category has the same absolute value, this eliminates the assumption that the category is ordinal or interval data.

AI used for this answer: Yes / **No**

If yes, how:

External references (links):
